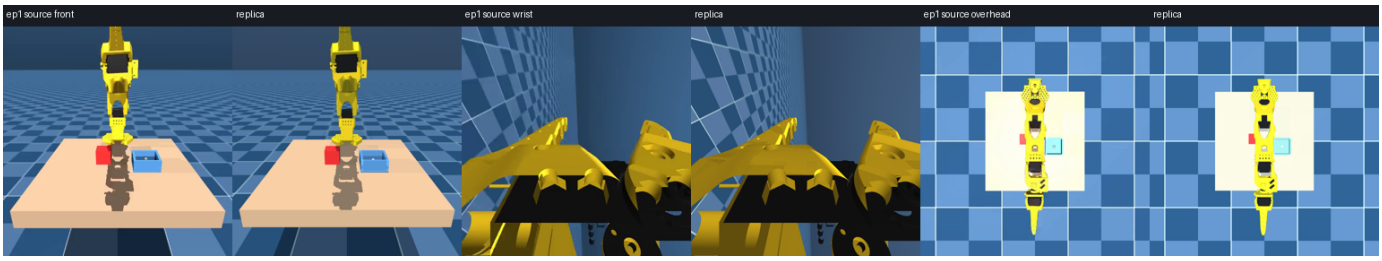
SO-101 MANIPULATION / 2026

SmolVLA on the SO-101

Evaluating SmolVLA manipulation on a simulated SO-101 arm



Source / replica pairs: front, wrist and overhead views

40k
archived fine-tuning updates

78.6%
lower held-out action MAE

200
evaluated policy rollouts

TASK

Evaluate how a fine-tuned vision-language-action policy behaves in a reconstructed robot environment, and identify the gaps between demonstration prediction and completed pick-and-place tasks.

IMPLEMENTATION

- Adapted a published MuJoCo replica with three camera views, six named joint channels, seeded resets and contact-based grasping.
- Integrated the archived 40k-step SmolVLA checkpoint and its saved processors; ran frozen inference on a 6 GB RTX 3060.
- Built stage-level evaluation, paired experiments, confidence intervals, per-seed physical traces and success/failure video evidence.

CLOSED-LOOP CONTROL

OBSERVE
3 RGB views
6 joint angles

>

PREDICT
SmolVLA
50 joint targets

>

EXECUTE
25 targets
MuJoCo physics

>

EVALUATE
Task stages
Stable placement

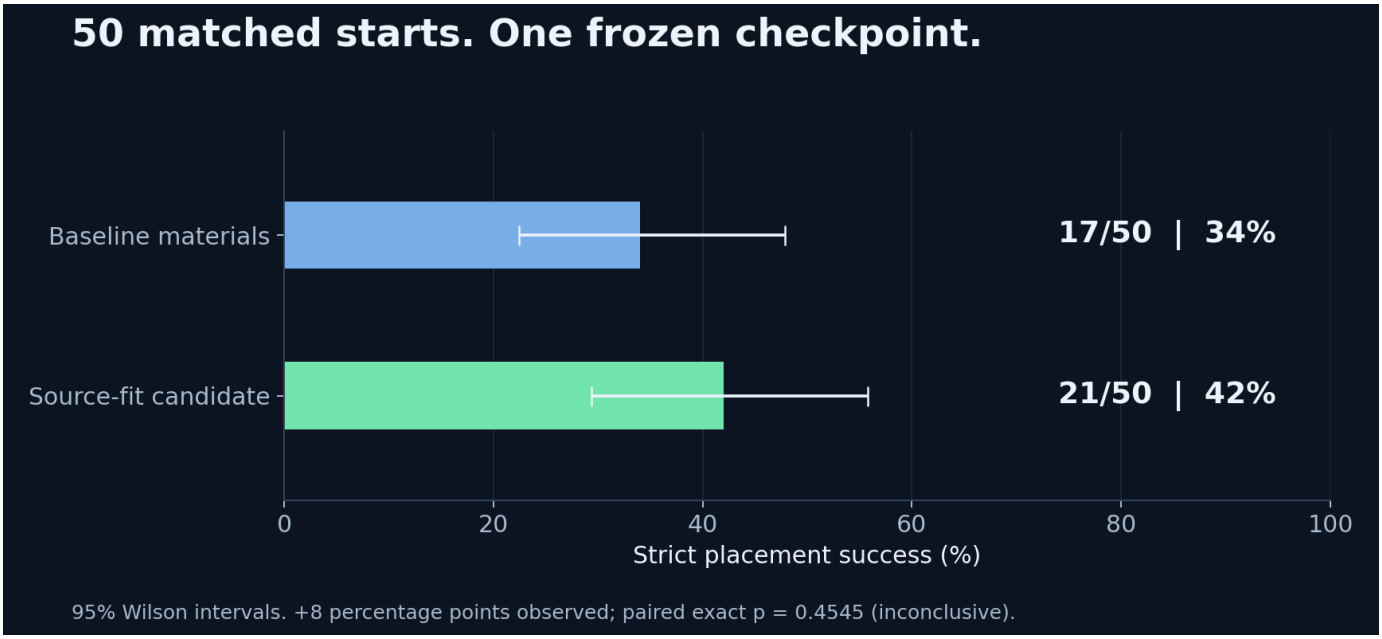
New observations after each executed chunk. Control period: 34 ms. Physics stepping is serialized; the simulator waits while the policy predicts.

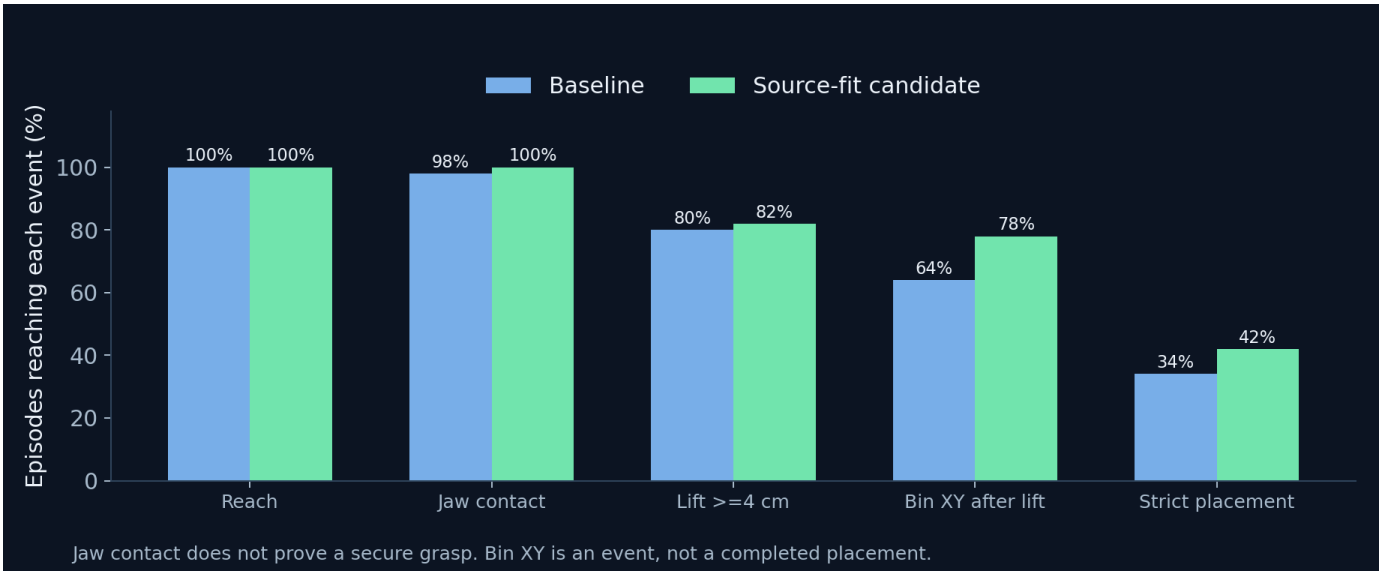
STACK **Python / PyTorch / LeRobot / MuJoCo / Gymnasium**

Training: Threadripper PRO / RTX 6000 Ada 48 GB. Inference: RTX 3060 6 GB. Offline action MAE uses 128 held-out trajectory anchors; it measures prediction error rather than task completion.

EVALUATION RESULTS

Same 40k weights / 50 paired starts / original anti-aliased rendering





FINDINGS

The source-fit candidate reached the cube in 50/50 trials, lifted it in 41/50 (82%) and completed strict placement in 21/50 (42%). Placement remains the main bottleneck.

Baseline placement was 17/50 (34%). The observed +8 percentage points are inconclusive: paired exact p = 0.4545. Candidate 95% Wilson interval: 29.4-55.8%.

A zero-tolerance containment defect undercounted settled cubes. Exact replays of 100 old traces verified a 0.1 mm contact-tolerance correction; it changes measurement, not actions.

Scope: Simulation-only evaluation in a published replica; original contact physics remain uncertain. Material selection used one source episode. No real-robot validation or statistically established material improvement is claimed.

FILES demo.mp4 / per-trial clips / JSON and CSV outcomes / source hashes

CREDITS Hugging Face SmolVLA/LeRobot; gpudad dataset; Rovo Labs scene (Apache-2.0).